Component diagram - MaleMail

View Layer (PySide6 GUI)

Main, Folder & Search UI
[FR-10, FR-11, FR-12]
[FR-13, OPT-01]

Email Viewer
(QWebEngineView)
[FR-08, FR-14, OPT-02]

Composer UI
[FR-05, FR-07, FR-08]
[FR-09, OPT-01, OPT-02]

Settings UI
[FR-09, OPT-01, OPT-03]

Calendar UI
[OPT-02]

Login UI
[FR-01, FR-04, OPT-01]

Controller Layer

MailController
(Handles Mail & Sync Events)
[NFR-01, NFR-03]

CalendarController
(Handles Calendar Events)
[NFR-01]

AuthController
(Handles Login & Auth Signals)
[NFR-01]

Model Layer (Python Core)

Inbound & Sync Service
[FR-06, FR-08, FR-10, FR-11, FR-12, OPT-01]

Outbound & MIME Service
[FR-05, FR-07, FR-08, FR-09, OPT-01, OPT-02]

Calendar & .ics Service
[OPT-02]

Spam Filter Engine
[OPT-03]

Auth & Session Manager
[FR-01, FR-02, FR-03, FR-04, OPT-01]

Local Data & Storage Controller
[FR-06, FR-08..13, OPT-01..03]

External Services & Protocols

External Mail Servers
(IMAP / SMTP)

Google Cloud & MS Entra ID
(OAuth2 & Calendar APIs)

Local Persistence Layer

Local File & Token Store
(os / pathlib / credentials.json)
[FR-02, FR-03, FR-08, OPT-01, OPT-02]

SQLite Database
(Emails, Flags, Signatures, Blacklist)
[FR-09..13, OPT-01..03]